

Autonomy, Clean Fuel and Digital Control Across Britain's Four Modes

Edition 2 in the UK Department for Transport series. Previous Decision Intelligence briefing: 'Decarbonise, Automate, Multiply' (May 2026).

Cycle: 17 September 2026 · Audience: DfT leadership team

· Horizon: to 2040

Read estimate: 7 min Executive Synthesis · 30 min full read

What you will learn

1. Why permission stopped being the constraint on all four modes in the same year
2. Which capacity reading moved against the department between May and September, and by how much
3. How the pathway choice got made without anyone taking the decision

Key takeaways

1. Rules landed in all four modes in 2026; the fleets they govern land in the 2030s
2. Battery supply is 31 GWh confirmed against roughly 130 GWh of 2035 demand, and one port was offered a 2039 grid connection
3. Four modes are separately solving the same unsolved problem, how to certify a system that decides for itself

Why This Matters Now

Three technology families now cut across every mode Britain runs: autonomy, clean fuel and digital control. In each of the four modes, 2026 was the year the rules arrived. Driverless passenger services became permissible in May. Britain authored the first international code for autonomous ships. Revenue support for sustainable aviation fuel became law in March. The regulator set a dated path for air taxis and for routine drone flight.

What did not arrive was the capacity to use any of it. Digital signalling is waiting on freight locomotives that are not being fitted. Battery manufacturing is running at a fraction of what the 2035 vehicle fleet needs. Ports have been told to wait until 2039 for a grid connection. Nobody publishes how many sustainable aviation fuel plants are actually being built.

The one thing to take into the next meeting: permission is no longer the binding constraint on any mode, so the department's 2040 exposure has moved from regulatory speed to industrial and assurance capacity, and those are owned by other people.

Executive Synthesis

Which decisions on your desk this week assume a build rate that no British supply chain has yet demonstrated?

The rail freight target needs 40% more capacity by 2040 and depends on fitting the Class 66 fleet, work that is currently paused; the 2035 vehicle fleet needs roughly 130 GWh of batteries against 31 GWh confirmed.

The May cycle asked which technologies would reach British transport by 2040. Four months later the more useful question is which of them Britain will be able to operate, and the answer separates sharply by mode. In every one of the four, 2026 delivered the rule and withheld the capacity. Driverless passenger services became permissible on 15 May under regulations made in April ([SI 2026/439](#), Apr 2026), sitting on an Act that is deliberately a skeleton ([the Automated Vehicles Act 2024](#), May 2024). The Maritime and Coastguard Agency led the drafting of the first international code for autonomous cargo ships, published on 1 July ([the Maritime and Coastguard Agency](#), Jul 2026), with the UK delegation described by government as instrumental ([DfT](#), Jun 2026). Revenue certainty for sustainable aviation fuel received Royal Assent on 5 March ([the Sustainable Aviation Fuel Act 2026](#), Mar 2026). The regulator holds a dated commitment to first commercial passenger air-taxi flights by the end of 2028 ([CAA CAP 3169](#), Sep 2025) and a phased path to routine drone operations beyond visual line of sight ([CAA CAP 3269](#), Jul 2026).

Against that, the capacity readings all moved the wrong way. Two confirmed gigafactories leave a 64 to 99 GWh shortfall against 2035 demand ([the UK Gigafactory Commission](#), Jan 2026). Public charge-point installations ran at 5,119 in the first half of 2026 against 13,281 in the whole of 2025 ([DfT official statistics](#), Aug 2026). Freight locomotive fitment, the thing digital signalling on the East Coast now waits on, is paused

pending a sector-wide design model ([Rail Engineer](#), Apr 2026). Ports told the department that grid connection lead times reach 15 years, with one offered 2039 ([DfT](#), Aug 2026). No government source states how many sustainable aviation fuel plants are under construction ([DfT](#), Jun 2026). This is not a story about technology maturity. It is a story about the distance between a rulebook and a fleet.

The May cycle framed the central choice as multi-pathway optionality against picking winners, and expected it to be settled at a fiscal event. It has been settled, but not by anyone choosing. On road, the mandate review is framed entirely around battery electric and plug-in hybrid vehicles and does not mention hydrogen ([DfT](#), Aug 2026); the £1 billion commercial vehicle package funds electric trucks, vans and depot charging with no hydrogen component ([DfT](#), Mar 2026); and the flagship hydrogen truck demonstrator closed after failing to convert expressions of interest into orders ([Trucking](#), Dec 2025). On aviation the department has explicitly kept several fuel pathways open ([DfT](#), Jun 2026) while the industry's own roadmap concedes hydrogen has slipped ([Sustainable Aviation](#), Jul 2026). Winners were picked by funding decisions and project failures, mode by mode, without a decision ever being minuted as such.

permission is no longer the binding constraint on any mode, so the department's 2040 exposure has moved from regulatory speed to industrial and assurance capacity, and those are owned by other people.

Underneath all four modes sits one unresolved question, and it is the reason the cross-modal theme is in this cycle at all. Every autonomy programme the department is now sponsoring depends on being able to certify a system whose behaviour is not deterministic. The statutory test put out for consultation in June is equivalence with a human: "A careful and competent human driver represents a higher standard than an average driver." ([DfT](#), Jun 2026). The peer-reviewed assessment of whether the department can apply that test finds no measurable engineering definition of the benchmark, no agreed method for quantifying acceptable risk, and no harmonised assessment method, certification process or post-market surveillance to apply it with ([Case Studies on Transport Policy](#), Feb 2026). The maritime code that Britain authored is non-mandatory until a mandatory version arrives in 2032 ([the Maritime and Coastguard Agency](#), Jul 2026). The drone concept of operations stops at 2028 ([CAA CAP 3269](#), Jul 2026). None of these is a gap in ambition. They are the same gap in assurance capability, appearing four times.

Two comparators are worth holding alongside this. Germany has reached the identical diagnosis on rail, that vehicle fitment rather than trackside works is the gating constraint, and has answered it with a funding directive of roughly 1.7 billion euros to 2030 covering up to 90% of first-of-class fitment, announced in German, “Mit unserer Förderrichtlinie machen wir die Züge fit für eine digitalisierte Schieneninfrastruktur.”, which reads in English as “With our funding directive we are making the trains fit for a digitalised rail infrastructure” ([the German federal transport ministry](#), Jul 2026). Norway has answered the marine fuel question from the demand side, requiring every company on its continental shelf to cut vessel emissions intensity from 2029 on a trajectory that runs to 2040, in Norwegian, “Regjeringen innfører krav for alle selskaper på norsk sokkel om å redusere klimagassutslippene fra skip som de bruker i olje- og gassvirksomheten fra 2029.”, which reads in English as “The government is introducing a requirement for all companies on the Norwegian shelf to reduce greenhouse gas emissions from the ships they use in oil and gas activity from 2029” ([the Norwegian Ministry of Climate and Environment](#), May 2026). Both are instrument choices Britain has not made. The UK has legislated permission in four modes and has not yet legislated, funded or mandated the conversion that permission was supposed to unlock.

Where this analysis could be wrong

Regulatory leadership may be worth more than this reading allows. Writing the standard others adopt is a real asset, and the maritime case is the strongest version of it: a code drafted under UK leadership becomes the rulebook every flag state works to from 2032 ([the Maritime and Coastguard Agency](#), Jul 2026). If British classification, insurance, assurance and software firms capture the services that regime creates, the absence of a British shipyard matters less than this brief implies. The thesis weakens if UK service-sector share of the autonomous maritime value chain can be shown to be growing.

The capacity numbers are not all the department's to move. Grid connection queues, battery plant economics and shipyard capacity sit with other departments, regulators and private capital. An argument that DfT is failing on capacity has to concede that the department's instruments are mostly demand-side. The honest form of the claim is narrower: the department is the only actor that can sequence its own permissions against the capacity that exists, and it has not been doing that.

Parts of this evidence base are thinner than a source count suggests. The strongest claim in the aviation theme rests on an absence, that no government source publishes a plant count, which is harder to defend than a positive finding. Two of the four cross-modal readings on cyber and satellite navigation are European rather than UK sources and are used as read-across ([ENISA](#), Jun 2026) ([EASA](#), Jul 2026). No current UK-specific source on vehicle-to-everything communication could be verified in window at any tier, so the connected half of connected and automated vehicles is carried here by a single peer-reviewed paper ([Case Studies on Transport Policy](#), Feb 2026) and should be treated as the weakest-evidenced part of the road theme.

The decisions that cannot be deferred this cycle

1. **Sequence permissions against capacity, in public.** The department controls when each permission regime goes live. It does not control gigafactories, grid connections or shipyards. Publishing the assumed build rate behind each regime is the one move that makes the mismatch visible before it becomes a delivery failure.

- 2. **Decide whether first-of-class rail fitment is a funded programme or a market failure.** Digital signalling, the 40% freight capacity target and the case for battery traction all resolve to the same unfunded question, and a peer administration has already priced an answer.
- 3. **Name an owner for cross-modal autonomy assurance.** Four modes are each building a safety case for non-deterministic systems, and the department's own research priorities do not list autonomy assurance as a theme.
- 4. **Treat the absence of a SAF plant count as a reportable failure.** A target expressed in plants under construction cannot be managed if nobody publishes the number, and the revenue certainty mechanism is now the instrument whose effect that number would measure.

Each is developed below, with a decision posture, in the four Strategic Implications.

Continuity ledger: what the May edition said and what happened

The May cycle left seven milestones to track. Six were testable and are graded here. The seventh could not be tested as written, because the decision point it named does not exist, and that is recorded as an overturned reading rather than quietly dropped.

What May claimed	Verdict	What happened
Automated passenger service regulations published, target Q1 2026	Strengthened	Delivered, one quarter later than the May reading expected and narrower than it assumed. SI 2026/439 was made on 20 April 2026 and came into force on 15 May 2026 (SI 2026/439 , Apr 2026), the day before the last cycle published. Bus-like services are covered in England, Wales and Scotland; taxi and private-hire-like services are England only, which the May cycle did not anticipate.
CAA legislative recommendations on eVTOL operations, due 2026, and VX4 flight testing in Q2 2026	Holding	The flight-test half is ahead of the reading. The VX4 completed a full two-way piloted transition at Kemble in April 2026, making Vertical the second developer worldwide to do so (Aviation International News , Apr 2026). The regulatory half has not moved visibly: the delivery model committing to initial commercial passenger flights by the end of 2028 remains at the version published in September 2025 (CAA CAP

3169, Sep 2025), and no 2026 legislative recommendation to the department could be verified.

2026 spending review pathway-narrowing decision on decarbonisation

Overturned

The premise was wrong. There was no 2026 spending review; the last multi-year settlement was Spending Review 2025. The decision the May cycle expected to be taken deliberately at a fiscal event has instead been taken by default and mode by mode, which is the subject of this cycle's first and fourth themes.

UK SAF plants under construction against the five-plant commitment by end-2026

Weakened

Not gradeable from the public record, which is itself the finding. The June 2026 announcement of a £219 million low carbon fuels fund names British Sugar at Wissington and a LanzaTech Humberside facility and gives no total (DfT, Jun 2026). No 2026 government source publishes an audited count of plants in construction.

UK gigafactory final-investment-decision pipeline through 2026 and 2027

Weakened

Materially worse than the May reading. The Gigafactory Commission reported two confirmed UK plants giving 31 GWh by 2030, at most 66 GWh with full site expansion, against roughly 130 GWh of 2035 demand, a shortfall of 64 to 99 GWh (the UK Gigafactory Commission, Jan 2026).

Second national e-scooter evaluation concludes, due 2026

Weakened

Slipping. The research protocol states the department expects the evaluation to report in 2026 (DfT, Sep 2025); as at this cycle's date no findings report has appeared. Rental trials now run to May 2028 and private e-scooters remain illegal on public roads, so the legal position for the 2030s is still open.

National mobility data platform design principles paper, target 30 June 2026

Holding

Delivered in a different shape. The April 2026 Transport Data Action Plan commits to a Transport Data Marketplace, to APIs as the expected sharing method and to a Transport Data Ontology (DfT, Apr 2026). A marketplace that links existing holdings is not the platform the May cycle described,

and the distinction decides who holds the data.

Audience Snapshots

Four lenses on the same evidence base. Each card surfaces the one question this cycle puts to that reader.

POLICY & REGULATION

Every mode now has its rule, and none of them binds the thing that is short

The shift: Driverless passenger permits in force from 15 May ([SI 2026/439](#), Apr 2026), the autonomous ships code published 1 July and non-mandatory until 2032 ([the Maritime and Coastguard Agency](#), Jul 2026), revenue certainty for aviation fuel law since March ([the Sustainable Aviation Fuel Act 2026](#), Mar 2026), and a dated regulatory path to air taxis by end-2028 ([CAA CAP 3169](#), Sep 2025).

The question to brief: Which of our four permission regimes would we delay by a year if delaying it bought a matching increase in the capacity to use it?

INFRASTRUCTURE & INVESTMENT

The queues are longer than the programmes that depend on them

The shift: Ports report grid connection waits of up to 15 years, one offered 2039 ([DfT](#), Aug 2026); charge-point installation halved year on year ([DfT official statistics](#), Aug 2026); and East Coast digital signalling now waits on freight locomotive fitment that is paused ([Rail Engineer](#), Apr 2026).

The question to brief: Where in our portfolio is a 2030 commitment resting on a connection, a fitment or a plant with a lead time longer than the time remaining?

DECARBONISATION & NET ZERO

The pathway question was answered by default, differently in each mode

The shift: The vehicle mandate review is framed on battery and plug-in hybrid with no mention of hydrogen ([DfT](#), Aug 2026), the hydrogen truck demonstrator closed ([Trucking](#), Dec 2025), aviation deliberately keeps multiple fuel pathways open ([DfT](#), Jun 2026), and marine alternative-fuel ordering fell about 40% year on year as the global carbon-pricing regime stalled ([DNV's Alternative Fuels Insight, reported by SAFETY4SEA](#), Apr 2026).

The question to brief: Are we content that the technology choices of the 2030s were made by which projects ran out of money first?

INNOVATION, INDUSTRIAL STRATEGY & SKILLS

Britain is buying the deployment it wrote the rules for

The shift: The first mainline battery fleet is an Alstom platform on a French contract built at Derby ([Railway Gazette](#), Sep 2026), read in Paris as an industrial export ([Economie Matin](#), Sep 2026); battery supply is 31 GWh against roughly 130 GWh of 2035 demand ([the UK Gigafactory Commission](#), Jan 2026); and 7% of science graduates entered manufacturing, with under-25 apprenticeship starts down 40% in a decade ([the House of Commons Transport Committee](#), May 2026).

The question to brief: On which of these technologies do we intend to hold the standard, the build, or neither, and has anyone written that down?

Themes

The cycle's signals are organised into six themes, ranked by impact on near-term decisions.

Immediate: changes the near-term plan, the structure or the proposition. **Near-Term:** changes position over the next twelve months. **Longer-Range:** a multi-year factor to track and revisit each cycle.

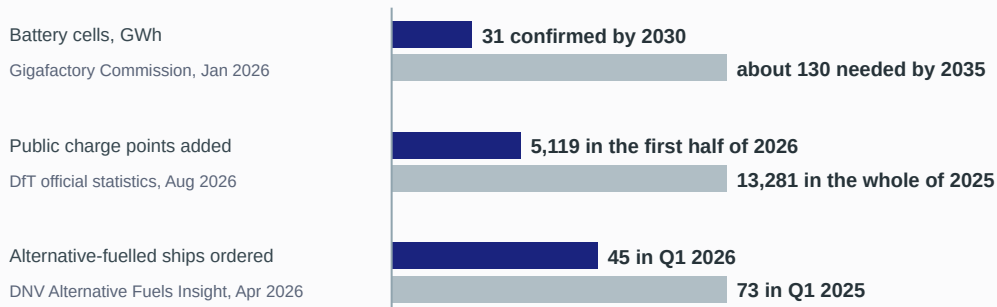
1. Road, where the permissions arrived before the industrial base

IMMEDIATE

Road is the mode where Britain moved first and has least left to decide about permission. The Automated Vehicles Act 2024 is deliberately a skeleton ([the Automated Vehicles Act 2024](#), May 2024), and the flesh arrived on 15 May 2026 when the permits regulations came into force ([SI 2026/439](#), Apr 2026). What the department now owns is not a regulatory question but

an industrial one, and the industrial readings have all moved against it since May. The vehicles that the permits allow will run on batteries the UK does not make in quantity, charged on infrastructure whose installation rate has halved.

What Britain has, against what the same policy assumes



Each pair is drawn to its own scale. The grey bar is the figure the policy is written to; the blue bar is what the public record currently shows.

- The permits regime is narrower than it is usually described. Bus-like services are covered across England, Wales and Scotland; taxi and private-hire-like services are England only ([SI 2026/439](#), Apr 2026). A national robotaxi market is not what has been authorised.
- Two confirmed gigafactories give 31 GWh by 2030 and at most 66 GWh with full site expansion, against roughly 130 GWh of demand by 2035, a shortfall of 64 to 99 GWh. The Commission's remedy list starts with attracting an anchor global manufacturer ([the UK Gigafactory Commission](#), Jan 2026).
- Public charge-point installation is decelerating. 5,119 devices were added in the first half of 2026 against 13,281 across the whole of 2025, and provision ranges from 345.1 devices per 100,000 people in London to 60.0 in Northern Ireland ([DfT official statistics](#), Aug 2026).
- The mandate review, open until 23 October 2026, is framed entirely around battery electric and plug-in hybrid vehicles and does not mention hydrogen cars or vans ([DfT](#), Aug 2026), while the Gigafactory Commission argues in the opposite direction, for cutting the 2030 target to 50 to 60% ([the UK Gigafactory Commission](#), Jan 2026).
- On freight the choice has already been made by spending. The £1 billion commercial vehicle package funds electric trucks, vans and depot charging with no hydrogen element ([DfT](#), Mar 2026), three months after the flagship hydrogen truck demonstrator closed having failed to convert more than 100 expressions of interest into binding orders ([Trucking](#), Dec 2025).

Early indicators

- **WEAK SIGNAL** **The government has tied its regulatory lead to one company.** A non-binding memorandum with Wayve commits government and a single domestic developer to collaborate on safety assurance, simulation testing and the move from pilots to commercial service ([Zag Daily](#), May 2026). It is intent rather than commitment, but it means the assurance methodology and the national champion are being developed by the same pair of hands.
- **WEAK SIGNAL** **Micromobility is still governed by a trial that keeps being extended.** Private e-scooters remain illegal, rental trials now run to May 2028, and the evaluation that was to inform legislation has not reported ([DfT](#), Sep 2025). The active travel strategy pushes the vehicle classification question into a separate powered mobility device review ([DfT](#), Jun 2026), so the legal position for the 2030s is being deferred rather than designed.

Counter-argument

Battery import dependence may simply be the efficient answer. Britain does not make its own semiconductors either, and a 64 to 99 GWh gap filled from European or Asian plants is a trade position rather than a failure, provided the cells arrive. The argument here is narrower than industrial sovereignty: rules of origin, the 2030 mandate trajectory and the depot-charging subsidy were all designed on the assumption of domestic supply, so the gap is a policy-coherence problem before it is an industrial one. The thesis weakens if the mandate review concludes with an import-based compliance route made explicit.

Decision link: Whether the department is regulating a market it expects Britain to supply, or one it expects Britain to import, and whether the mandate review says which.

2. Rail, where the digital upgrade is waiting on the trains

NEAR-TERM

Rail is the clearest case in this cycle of a technology programme whose remaining risk has moved entirely into one unfunded activity. The trackside half of digital signalling on the East Coast is broadly working: the Northern City Line runs under full ETCS with lineside signals removed, and passenger fleets are fitted or in fitment. The programme now waits on freight locomotives, and that work is paused pending a sector-wide design model, with departmental funding extending only to a defined configuration state ([Rail Engineer](#), Apr 2026). Everything else in the mode, including a new statutory freight target, rests on that.

- Government has set a rail freight target of at least 40% more capacity by 2040, with 75% growth by 2050, and made freight growth an explicit part of Great British Railways' mission: "Great British Railways will have a clear mission to help grow our economy by moving more of

the goods British businesses rely on.” (DfT, Sep 2026). The path capacity to carry it is what ETCS is supposed to release.

- The Railways Bill creates that guiding mind and consolidates retail and passenger data powers, but contains no technology mandate, no digital-signalling duty and no rolling-stock fitment obligation (DfT, Jun 2026). Fitment economics therefore sit outside the statutory regime entirely.
- Germany reached the same diagnosis and answered it with money. Its funding directive allocates roughly 1.7 billion euros to 2030 for vehicle fitment, covering up to 60% of series installation and up to 90% of first-of-class work, announced in German, “Mit unserer Förderrichtlinie machen wir die Züge fit für eine digitalisierte Schieneninfrastruktur.”, which reads in English as “With our funding directive we are making the trains fit for a digitalised rail infrastructure” (the German federal transport ministry, Jul 2026).
- Britain's first mainline battery fleet is contracted: 29 five-car units for TransPennine Express at about £1 billion including eight years of maintenance, with charging at Hull, Scarborough and Saltburn, delivery from 2032 and entry into service in winter 2034 (Railway Gazette, Sep 2026). Manufacture at Derby starts in 2028 (Alstom, Sep 2026), and the French business press reads the same contract as an export win, in French, “L'usine historique de Derby Litchurch Lane produira 29 rames dès 2032 pour le nord de l'Angleterre.”, which reads in English as “The historic Derby Litchurch Lane works will produce 29 trainsets from 2032 for the north of England” (Economie Matin, Sep 2026).
- The industry's own research warns against generalising from that order. Battery-only locomotives have a role in shunting and very short trips and are not a substitute for mainline electrification; “With their greater power and acceleration electric locomotives (and bi-mode and hybrid locomotives operating under electric power) can haul heavier and longer trains whilst delivering faster climbing, higher average speeds, and more available train paths for both freight and passenger services.” (the Railway Industry Association, Jul 2026). The argument is about path capacity at least as much as carbon.

Early indicators

- **WEAK SIGNAL** **The barriers to rail technology adoption were designed in, and the people fixing them say so.** GBR's innovation arm names three obstacles, commercial fragmentation, contractual and regulatory barriers built into the Network Code and the safety validation regime, and insufficient in-house frontier expertise (International Railway Journal, Apr 2026). Its managing director previously led the East Coast digital programme, so the fitment experience is visibly informing the diagnosis.
- **WEAK SIGNAL** **The data problem is the same one, one layer down.** A peer-reviewed survey of AI predictive maintenance for rail infrastructure finds the constraint has moved from model capability to fragmented datasets, poor generalisation between networks, uninterpretable models in a safety-critical regime, and disconnection from asset management platforms (Sensors, Jan 2026). Results from one network are not evidence about another.

Counter-argument

Freight fitment may be the right thing to pause. Class 66 locomotives are old, a first-of-class programme on a legacy fleet is the most expensive way to buy ETCS capability, and waiting for a sector-wide design model may cost less than fitting twice. That is a serious argument and it is why the claim here is about synchronisation rather than speed: a 2040 capacity target and a paused enabling programme cannot both be real. The thesis weakens if a funded fitment mechanism is announced with a completion date inside the target's horizon.

Decision link: Whether first-of-class freight fitment is funded as a programme, on the German model or another, or the 2040 freight capacity target is restated to match what unfunded fitment will deliver.

3. Maritime, where Britain wrote the rulebook for a fleet it does not build

LONGER-RANGE

Maritime is the mode where British regulatory leadership is most real and most detached from British industrial capacity. The Maritime and Coastguard Agency led the drafting of the first international regime for remotely operated and autonomous cargo ships, published on 1 July 2026 ([the Maritime and Coastguard Agency](#), Jul 2026), with government describing the UK delegation as instrumental ([DfT](#), Jun 2026). It is non-mandatory, and a mandatory version is scheduled for 2032. Everything commercially consequential about it therefore happens in a window Britain has six years to position for, during which the global rules on what those ships burn remain unsettled.

- The Agency frames publication as an enabling act: “Publishing the MASS Code is a major step towards delivering clearer, stronger, safer regulations that reduce risk and support innovative technology.” ([the Maritime and Coastguard Agency](#), Jul 2026). The experience-building phase that follows is where the operating practice, and the assurance services around it, get defined.
- The domestic regime bites sooner than the international one. Workboat Code Edition 3 consolidates the small commercial vessel codes and includes what the Agency calls the world's first maritime safety legislation for remotely operated unmanned vessels and for workboats on alternative fuels, with the first cohort required to transition by 13 December 2026 ([the Maritime and Coastguard Agency](#), Feb 2026).
- The fuel question has stalled at global level. The extraordinary session in October 2025 voted 57 to 49 with 21 abstentions to adjourn adoption of the Net-Zero Framework for twelve months, and entry into force is now unlikely before 2028 ([Hill Dickinson](#), Oct 2025). Talks resume in October 2026, a month after this cycle publishes.

- Owners have responded by not ordering. Alternative-fuelled vessel orders fell to 45 in the first quarter of 2026 from 73 a year earlier, with deliveries dominated by LNG and methanol and ammonia and hydrogen newbuilds marginal ([DNV's Alternative Fuels Insight](#), reported by [SAFETY4SEA](#), Apr 2026). The fleet arriving before 2030 is already largely fixed.
- Where the UK does control the asset, the constraint is the grid. Ports reported connection lead times of up to 15 years, one told its connection would not occur until 2039, and shore power standing charges between £157,093 and £413,749 a year. The department's own summary records that "Shore power is seen as a mature technology and has been proven to reduce at-berth emissions if the electricity is renewably sourced." ([DfT](#), Aug 2026).

Early indicators

- **WEAK SIGNAL** **Norway is regulating the charterer, not the ship.** Every company operating on the Norwegian shelf must cut the emissions intensity of the vessels it uses from 2029, phasing to 40% by 2038 to 2040: in Norwegian, "Regjeringen innfører krav for alle selskaper på norsk sokkel om å redusere klimagassutslippene fra skip som de bruker i olje- og gassvirksomheten fra 2029.", which reads in English as "The government is introducing a requirement for all companies on the Norwegian shelf to reduce greenhouse gas emissions from the ships they use in oil and gas activity from 2029" ([the Norwegian Ministry of Climate and Environment](#), May 2026). It is demand-side, it runs to this cycle's horizon, and it applies to a support fleet that also works British waters.
- **WEAK SIGNAL** **Port digital twins have outrun the evidence that they work.** Benchmarking seven European port twin initiatives finds the architectures converging on predictive and prescriptive decision support, but no documented synchronisation, latency or model-performance reporting ([Results in Engineering](#), Jun 2026). A port authority cannot currently tell whether a twin is fit to carry an operational or safety decision, which is the same assurance gap the fifth theme describes, one layer down.
- **WEAK SIGNAL** **Rotterdam is fixing the interfaces UK freight will have to meet.** The 2050 port vision commits to moving hydrogen and hydrogen carriers by pipeline wherever possible and to sensor-level digital integration across cargo, transport and infrastructure ([the Port of Rotterdam](#), Dec 2025). As the dominant transshipment node for UK-bound freight, its choices constrain what British ports can interoperate with.

Counter-argument

Authoring the standard may be worth more than building the ships. If British classification, insurance, assurance, software and remote-operations services capture the market the 2032 regime creates, the absence of a domestic shipyard is close to irrelevant, and the UK will have converted regulatory effort into service revenue at very low capital cost. This brief does not dismiss that reading; it notes that nothing in the public record measures it. The thesis weakens if UK service-sector share of the autonomous maritime value chain can be shown, and the department should want that number.

Decision link: Whether the UK's return on the autonomous shipping code is measured in services captured by 2032, and if so who is measuring it.

4. Aviation, where the fuel pathway became law before the fuel

NEAR-TERM

Aviation has the most complete policy architecture of the four modes and the least visible physical delivery. Revenue certainty became law on 5 March 2026, letting the Secretary of State direct contracts that pay producers when the strike price exceeds the market reference price, with the power expiring after ten years ([the Sustainable Aviation Fuel Act 2026](#), Mar 2026). A mandate rising from 2% in 2025 to 10% by 2030 and 22% by 2040 guarantees the demand ([DfT](#), Jun 2026). What no government source states is how much plant is being built to meet it.

The rules land in 2026. The things they govern land in the 2030s.

Mar 2026	●	Sustainable Aviation Fuel Act receives Royal Assent
May 2026	●	Driverless passenger service permits come into force
Jul 2026	●	Autonomous ships code published, non-mandatory
2028	●	First air-taxi passenger flights targeted; battery-train build starts
2032	●	Battery trains delivered; autonomous ships code becomes mandatory
2034	●	Those battery trains enter passenger service
2039	●	Grid connection date offered to one UK port
2040	●	Rail freight capacity +40%; aviation fuel mandate reaches 22%

Dates as printed by the instrument or contract that sets them. The gap between the first three entries and the rest is the subject of this cycle.

- The June 2026 announcement of a £219 million low carbon fuels fund, with £93 million available immediately, names British Sugar at Wissington and a LanzaTech facility on Humberside and gives no total count of plants in construction ([DfT](#), Jun 2026). It does not mention the Act passed three months earlier.
- The department has deliberately kept several fuel pathways open, restating that the power-to-liquid obligation starts in 2028 and that HEFA-derived supply falls as a share of eligible fuel to 2040 ([DfT](#), Jun 2026). That is the opposite of the pick-a-winner posture taken on road vehicles.
- The industry's own roadmap concedes that the other pathway has slipped: "The entry into service timeline for hydrogen narrow body aircraft has been adjusted to reflect the complex engineering timelines required to mature novel technology, and to align with more conservative assessments of hydrogen production and infrastructure deployment schedules." ([Sustainable Aviation](#), Jul 2026). Its central scenario has fuels, not airframes, carrying most of the decarbonisation.

- Independent reporting puts the same slippage on the manufacturer side, with full certification of one hydrogen-electric powertrain pushed back about two years to a 2027 fuel cell target after a workforce halving ([Runway Girl Network](#), Feb 2026). German national research is still working the cryogenic fuel system between tank and turbine ([the German Aerospace Center](#), Mar 2026), which is the component-level reason the dates keep moving.
- Air taxis and drones are further ahead on regulation than on airspace. The regulator holds an end-2028 commitment for first commercial passenger eVTOL flights ([CAA CAP 3169](#), Sep 2025) and a phased path to routine beyond-visual-line-of-sight operation whose own horizon stops at 2028 ([CAA CAP 3269](#), Jul 2026), while airspace redesign has been centralised into a single service that is London-first and funded by a new charge ([the CAA](#), undated).

Early indicators

- **WEAK SIGNAL** **The aircraft is doing better than the fuel.** The VX4 completed a full two-way piloted transition at Kemble in April 2026, the second developer worldwide to do so ([Aviation International News](#), Apr 2026). Airframe progress is real and is not the constraint; the constraint is airspace capacity and fuel supply, neither of which a manufacturer can solve.
- **WEAK SIGNAL** **Centralising airspace design concentrates the risk as well as the authority.** Airspace modernisation was blocked because no single party had the mandate or funding to redesign shared airspace; the answer routes it through one licensed provider, initially for London ([the CAA](#), undated). That is a faster model and a single point of failure, and wider rollout is conditional.

Counter-argument

The absence of a plant count may mean nothing. Final investment decisions are commercially sensitive, the revenue certainty mechanism only received Royal Assent in March and its first contracts would not yet show as construction, and a mandate that guarantees demand to 2040 is precisely the instrument designed to pull plant through. Judging the policy on a missing statistic six months in is premature. The claim here is therefore about management rather than failure: a target expressed in plants under construction cannot be steered if the number is not published, and the department set that target itself.

Decision link: Whether to publish a plant-by-plant construction count against the mandate trajectory, and whether aviation's multi-pathway posture survives contact with the single-pathway posture taken on road.

5. The assurance and data spine beneath all four modes

IMMEDIATE

The four modes look like four programmes and are increasingly one. Each is building a case that a system which decides for itself is safe enough to authorise, each depends on data it does not own, and each assumes positioning and timing signals that are being actively degraded. This theme exists because that layer has no departmental owner, and because the department's own statement of research priorities does not name autonomy assurance, cyber security or positioning resilience among its themes ([DfT](#), Jul 2026).

- The statutory safety test for self-driving is human equivalence, and the consultation is explicit about where the bar sits: "A careful and competent human driver represents a higher standard than an average driver." ([DfT](#), Jun 2026). That is a legal standard awaiting an engineering definition.
- Peer-reviewed assessment finds the UK has no measurable engineering definition of that benchmark, no agreed method for quantifying acceptable risk, and no harmonised assessment method, formal certification process or post-market surveillance ([Case Studies on Transport Policy](#), Feb 2026). The standards most often cited address deterministic systems rather than learned behaviour.
- The data answer is a marketplace rather than a platform. The April 2026 action plan commits that "We will develop a Transport Data Marketplace, exploring the best way to bring in and link to existing marketplaces, creating a single port of call for transport data." ([DfT](#), Apr 2026), with APIs as the expected sharing method and an ontology to link the system's parts. The integrated transport strategy pairs it with tools for city regions to run their own contactless ticketing by 2030 ([DfT](#), Apr 2026).
- Transport is inside the critical-sector cyber regime and Parliament has judged the 2018 regime inadequate, with the Bill widening duties and regulator powers ([the House of Commons Library](#), May 2026). The European crisis exercise this year ran a simulated scenario of simultaneous coordinated attack on maritime and rail infrastructure ([ENISA](#), Jun 2026), which is a rehearsal rather than an incident but is the shape of the risk.
- Positioning and timing are being treated as infrastructure for the first time, with a £155 million package covering terrestrial eLoran, a national timing centre and interference monitoring ([Inside GNSS](#), Nov 2025), while the European aviation regulator reports jamming and spoofing growing in severity and sophistication ([EASA](#), Jul 2026).

Early indicators

- **WEAK SIGNAL** **The workforce that has to operate all of this is contracting at the entry point.** Only 7% of science graduates entered manufacturing in 2021/22 and apprenticeship starts for under-25s have fallen 40% in a decade ([the House of Commons Transport Committee](#), May 2026). Every theme in this cycle assumes people who can assure, fit, operate and maintain systems that do not exist yet.
- **WEAK SIGNAL** **Assurance capacity is the constraint the modes will hit in the same year.** Maritime's experience-building phase, the drone concept of operations and the self-driving authorisation regime all mature between 2027 and 2032 ([the Maritime and Coastguard Agency](#), Jul 2026) ([CAA CAP 3269](#), Jul 2026) ([SI 2026/439](#), Apr 2026). They will compete for the same small pool of people able to certify non-deterministic systems.

Counter-argument

Assurance may not need to be centralised. Each mode has a regulator with deep domain knowledge, and a cross-modal assurance function could easily become a coordination layer that slows four programmes without improving any of them. The honest version of this theme is not that the department should build a central body, but that four regulators are independently discovering the same unsolved problem and none of them owns it. The thesis weakens if any one of the four publishes a quantified method for demonstrating that a non-deterministic system meets its statutory safety test, because the others will adopt it.

Decision link: Who owns the question of how a non-deterministic system is certified, given that four modes are now answering it separately and the department's research priorities do not list it.

Strategic Implications

Four decisions turn this cycle's evidence into the 2027 business plan. Two are decisions in the strict sense, one is preparation the department can start without a decision, and one is a measurement it does not currently take.

SI 1: Publish the build rate each permission regime assumes

Four permission regimes are now live or dated, and each was designed against an implicit assumption about how fast Britain can build the thing it permits. The road mandate assumes a cell supply that is 31 GWh confirmed against roughly 130 GWh of 2035 demand ([the UK Gigafactory Commission](#), Jan 2026). The freight capacity target assumes locomotive fitment that is paused ([Rail Engineer](#), Apr 2026) ([DfT](#), Sep 2026). The aviation mandate assumes plant nobody counts ([DfT](#), Jun 2026). None of these assumptions is written down in a form that lets anyone test it, which is why each is failing quietly rather than visibly. The department cannot build gigafactories or shipyards, but it is the only actor that can state what its own instruments presuppose.

Action: For each of the four permission regimes, publish the build or fitment rate it assumes and the current observed rate, in the 2027 business plan. Where the two diverge, say whether the regime or the assumption is being changed.

DECIDE

Draws on Themes 1, 2 and 4

SI 2: Resolve first-of-class rail fitment as a funded question

Three separate commitments now rest on the same paused activity. Digital signalling completion, the 40% freight capacity target by 2040 and the case for discontinuous electrification all require freight locomotives to be fitted, and the Railways Bill creates a guiding mind with no fitment obligation ([DfT](#), Jun 2026) ([DfT](#), Sep 2026) ([Rail Engineer](#), Apr 2026). A peer administration has already priced this problem and concluded the state must carry it, at up to 90% of first-of-class cost ([the German federal transport ministry](#), Jul 2026). Britain has not decided whether it agrees, and the absence of a decision is currently being read as a decision not to fund.

Action: Bring a funded fitment option to the next fiscal event with a completion date inside the 2040 freight target's horizon, or restate the target to match what unfunded fitment will deliver. Do not carry both.

DECIDE

Draws on Theme 2

SI 3: Give cross-modal autonomy assurance an owner

Road, maritime and aviation are each building a method for demonstrating that a system which decides for itself is safe enough to authorise, on timetables that mature together between 2027 and 2032 ([SI 2026/439](#), Apr 2026) ([the Maritime and Coastguard Agency](#), Jul 2026) ([CAA CAP 3269](#), Jul 2026). The statutory road test is equivalence with a careful and competent human driver ([DfT](#), Jun 2026), and the peer-reviewed finding is that no method, certification process or post-market surveillance exists to apply it ([Case Studies on Transport Policy](#), Feb 2026). The department's own research priorities do not list autonomy assurance as a theme ([DfT](#), Jul 2026). This does not need a new central body, which would probably slow all four. It needs one named owner of the question and a shared method the four regulators can adopt.

Action: Name a senior owner for cross-modal autonomy assurance before the statement of safety principles is finalised, and commission the quantified method the statutory test currently lacks.

PREPARE

Draws on Theme 5

SI 4: Measure the return on regulatory leadership

The strongest counter-argument to this cycle is that authoring standards is worth more than building hardware, and that the UK is converting regulatory effort into service revenue at low capital cost. It may be right. The maritime code the UK led is the test case, with a non-mandatory phase running to a mandatory regime in 2032 ([the Maritime and Coastguard Agency](#), Jul 2026) ([DfT](#), Jun 2026). Nothing in the public record measures whether British classification, insurance, assurance or remote-operations firms are capturing that market. Without the number, the department cannot tell whether its own best argument is true.

Action: Commission a baseline measure of UK service-sector share of the autonomous maritime value chain, and repeat it annually to 2032, so that the next cycle can grade regulatory leadership on evidence rather than assertion.

MONITOR

Draws on Themes 3 and 5

Scenario Matrix

The two axes are the two things this cycle says are unresolved, and neither is a forecast of demand or technology. The horizontal axis is where the capacity to deploy comes from. The vertical axis is whether the four modes converge on one way of assuring a system that decides for itself. The permission regimes are assumed in all four quadrants, because they already exist.

WHERE DEPLOYMENT CAPACITY COMES FROM

Standard and factory

The fitment question is funded, the mandate review lands with an industrial answer, and one of the four regulators publishes a quantified method for certifying non-deterministic systems that the others adopt. Britain holds both the rulebook and a meaningful share of the build. This is the scenario the 2026 legislative programme was implicitly written for, and it is the one that requires decisions the department has not yet taken. It is not the most likely, and saying so is the point of the matrix.

Early indicators: A funded first-of-class rail fitment mechanism with a completion date; The mandate review concluding with a stated domestic supply assumption; Any one regulator publishing a quantified autonomy assurance method

The standards economy

Assurance converges, and Britain exports it. The hardware is French, Korean, Chinese or American; the certification, classification, insurance, software and remote-operations services around it are British. The maritime code is the template and the 2032 mandatory regime is the revenue event. On present evidence this is the most plausible of the four, and it is a perfectly respectable outcome, but it is currently being arrived at by accident rather than chosen, which means nobody is building the service capacity it depends on.

Early indicators: UK firms visibly winning MASS assurance and remote-operations work; Growth in UK certification and classification employment; Continued import share in battery, rolling stock and vessel procurement

Four separate answers

Industrial capacity improves, through the mandate review, a fitment settlement or an anchor manufacturer, but each mode solves autonomy assurance on its own terms. Four incompatible methods emerge, each defensible

MODE-BY-MODE, BRITISH BUILD ¹⁵

on operators working across modes and on the small pool of people who can certify any of it. Britain builds, and pays for the same assurance problem four times.

Early indicators: An anchor battery manufacturer committing to the UK; Separate assurance methods emerging from CAA, MCA and DfT without cross-reference; Assurance skills shortages appearing simultaneously in two or more modes

Permission without capacity

The regimes go live on schedule and the fleets do not follow. Battery supply stays short, fitment stays paused, ports stay in the connection queue, and no plant count is published. Each regulator

MODE-BY-MODE, IMPORTED PLATFORMS

method under delivery pressure. Britain becomes the jurisdiction where the technology is legal first and deployed late, and the industrial and service returns accrue elsewhere. Nothing in the current evidence excludes this, which is why it is stated plainly rather than softened.

Early indicators: The 2040 freight target restated or quietly dropped; A second year with no published SAF plant count; Charge-point installation falling again in the second half of 2026

What We Are Not Planning For

PROPULSION

We are not planning for hydrogen to carry a material share of UK transport by 2040

Three independent lines of evidence point the same way: the flagship hydrogen truck demonstrator closed for want of orders ([Trucking](#), Dec 2025), the road mandate review does not mention hydrogen at all ([DfT](#), Aug 2026), and the aviation industry's own roadmap has pushed hydrogen narrow-body entry into service later than it assumed in 2023 ([Sustainable Aviation](#), Jul 2026), with the cryogenic fuel system still at rig stage ([the German Aerospace Center](#), Mar 2026). We treat hydrogen as a niche for specific duty cycles inside this horizon, not as a pathway.

↺ **Reinstated if** a funded UK hydrogen corridor reaches financial close, or the mandate review reopens hydrogen for light duty.

AUTONOMY AT SCALE

We are not planning for driverless services to be a national market before the 2030s

What came into force in May is a permit scheme, not a market. Taxi and private-hire-like services are authorised in England only, permits run for a maximum of five years and are individually granted ([SI 2026/439](#), Apr 2026), and the statutory safety test is still out for consultation ([DfT](#), Jun 2026). We model city-scale permitted services through the late 2020s and treat national coverage as a 2030s question.

↺ **Reinstated if** permits are granted at a rate implying multi-city coverage, or the safety principles are finalised with a quantified method attached.

GLOBAL FUEL PRICING

We are not planning for a global shipping carbon price to be in force before 2028

Adoption of the Net-Zero Framework was adjourned for twelve months on a divided vote and entry into force is now unlikely before 2028 ([Hill Dickinson](#), Oct 2025), with talks resuming in October 2026. Owners have already responded by slowing alternative-fuel ordering ([DNV's Alternative Fuels Insight](#), reported by [SAFETY4SEA](#), Apr 2026). We plan on regional instruments rather than a global price, and on a pre-2030 fleet that is already largely fixed.

↻ **Reinstated if** the reconvened session adopts the instrument with an entry-into-force date, or a coalition of flag states adopts an equivalent regional measure.

Questions For The Next Strategy Meeting

Five questions this cycle's evidence should put on the agenda.

- 01 For each of our four permission regimes, what build or fitment rate did we assume when we designed it, and what is the observed rate now ([the UK Gigafactory Commission](#), Jan 2026) ([Rail Engineer](#), Apr 2026) ([DfT](#), Jun 2026)?
- 02 The 2040 rail freight target needs 40% more capacity ([DfT](#), Sep 2026) and the fitment that releases it is paused ([Rail Engineer](#), Apr 2026). Which of the two are we changing?
- 03 Who in this department can tell a minister that a self-driving system meets the statutory test, and what method would they use ([DfT](#), Jun 2026) ([Case Studies on Transport Policy](#), Feb 2026)?
- 04 We led the drafting of the autonomous shipping code ([DfT](#), Jun 2026). What are we measuring to know whether that was worth doing?

05 The pathway choice this department expected to make deliberately has been made mode by mode by funding decisions and project closures (DfT, Aug 2026) (DfT, Mar 2026) (Trucking, Dec 2025) (DfT, Jun 2026). Do we endorse the result, and if so will we say so?

Source Confidence Register

This briefing draws on 49 verified sources, gathered under a declared twelve-month window (publications from September 2025 onward), with 2 structural anchors outside it. 5 are in a language other than English and were read in the original, because on this topic the translated version arrives late and loses the numbers.

Source tiers: **Tier 1**, governments, parliaments, regulators, courts and intergovernmental bodies. **Tier 2**, national audit offices, think-tanks, academic institutes and major professional firms. **Tier 3**, quality journalism and specialist trade press. **Tier 4**, vendor and practitioner sources, used only as directional corroboration. This cycle uses one Tier 4 source, Alstom, which anchors no claim.

Theme 1: Road, where the permissions arrived before the industrial base

Source	Tier	Date	Key claim used
Department for Transport (GOV.UK official statistics)	Tier 1	Aug 2026	At 1 July 2026 the UK had 121,171 public charging devices, of which 28,887 were rapid or above, averaging 174.9 per 100,000 population, ranging from 345.1 in London to 60.0 in Northern Ireland; only 5,119 devices were added in the first half of 2026 against 13,281 across the whole of 2025.
Department for Transport (GOV.UK)	Tier 1	Aug 2026	DfT has reopened the Vehicle Emissions Trading Schemes Order for review, seeking views on the ZEV trajectory, yearly headline targets and compliance flexibilities, framing the review entirely around battery electric and plug-in hybrid vehicles, restating the 2030 and 2035 phase-out dates, and making no reference to hydrogen fuel cell cars or vans; the consultation closes 23 October 2026.

Source	Tier	Date	Key claim used
Department for Transport (GOV.UK)	Tier 1	Jun 2026	CWIS3 commits that by 2030 a national walking and cycling network will be accessible through standard route mapping websites and apps in the same way as the road network, improves data standards for authorities to track network development, and defers powered mobility devices to a separate review.
UK Government / The National Archives (legislation.gov.uk)	Tier 1	Apr 2026	The Regulations set the procedural requirements for the automated passenger services permitting scheme under the Automated Vehicles Act 2024, covering bus-like services in England, Wales and Scotland and taxi and private-hire-like services in England; made 20th April 2026, laid before Parliament 23rd April 2026, coming into force 15th May 2026.
Department for Transport / Office for Zero Emission Vehicles (GOV.UK)	Tier 1	Mar 2026	The government announced £1 billion for commercial vehicle decarbonisation: a Zero Emission Truck Grant worth up to £81,000 per vehicle covering up to 40% of cost, a Zero Emission Van Grant of up to £5,000, and a £170 million Depot Charging Scheme covering up to 70% of installation costs.
Department for Transport (GOV.UK)	Tier 1	Sep 2025	DfT set out the design of a second national evaluation of England's rental e-scooter trials, states it expects the evaluation to report in 2026, records 18 live trials running to May 2028, and confirms that use of private e-scooters on public roads remains illegal.
UK Parliament / The National Archives (legislation.gov.uk)	Tier 1	May 2024	The Act creates the UK statutory framework for self-driving vehicles, covering authorisation, the authorised self-driving entity, licensing of no-user-in-charge operators, a Secretary of State Statement of Safety Principles, marketing restrictions on misleading autonomy claims, incident investigation, and permits for automated passenger services.

Source	Tier	Date	Key claim used
Case Studies on Transport Policy (Elsevier), Shah, Davies, Bastien and Debnath	Tier 2	Feb 2026	The paper finds three capability gaps in the UK connected and automated vehicle regime: the statutory careful and competent driver benchmark has no measurable engineering definition, there is no agreed method for quantifying acceptable risk under Positive Risk Balance, and the UK lacks a harmonised assessment framework, formal certification process or post-market surveillance for in-use monitoring.
The Faraday Institution (UK Gigafactory Commission)	Tier 2	Jan 2026	Only two UK gigafactories are confirmed, giving 31 GWh by 2030 rising to at most 66 GWh with full site expansion, against 2035 demand of about 130 GWh, leaving a shortfall of 64 to 99 GWh; the Commission sets out ten priority actions including attracting an anchor global OEM and resetting the ZEV mandate 2030 target to 50 to 60%.
Zag Daily	Tier 3	May 2026	The UK government signed a non-binding Memorandum of Understanding with London-based Wayve covering collaboration on safety assurance, simulation testing and the transition of pilot schemes to commercially operated services, with Wayve sharing operational insight with government and regulators to inform future regulation.
Trucking magazine	Tier 3	Dec 2025	HyHAUL, the DfT and Innovate UK-backed hydrogen HGV demonstrator that aimed to deploy 30 fuel-cell trucks with refuelling along the M4 corridor, ceased after failing to convert more than 100 expressions of interest into binding orders within funding deadlines, despite completing four of its five work packages.

Theme 2: Rail, where the digital upgrade is waiting on the trains

Source	Tier	Date	Key claim used
Department for Transport (GOV.UK)	Tier 1	Sep 2026	Government has set a rail freight target to increase capacity by at least 40% by 2040, with an ambition of 75% growth by 2050, which it says could move an additional 15 billion pounds of goods annually, and has made growing freight an explicit part of GBR's mission.
Bundesministerium für Verkehr (BMV), Germany <i>(in German)</i>	Tier 1	Jul 2026	The German federal transport ministry has opened a funding directive to retrofit existing rolling stock with ERTMS and ETCS, allocating roughly 1.7 billion euros to 2030 for nationwide vehicle fitment plus a further 482 million euros for the Digitaler Knoten Stuttgart, covering up to 60% of series installation costs and up to 90% of first-of-class prototype fitment.
Department for Transport (GOV.UK)	Tier 1	Jun 2026	The Railways Bill creates Great British Railways as a publicly owned body duty-bound to promote high standards of performance, enables consolidation of ticket retail currently spread across 14 train companies into a single GBR app and website, gives the passenger watchdog powers to collect performance data, and requires GBR to produce an integrated five-year business plan subject to regulator approval.
Railway Industry Association, with Rail Freight Group and research by the University of Birmingham	Tier 2	Jul 2026	The report concludes that battery-only locomotives have a role only in shunting and very short trips and are not a substitute for mainline electrification, that electric locomotives are the best long-term option for freight with electro-diesel as an interim, and that targeted infill electrification is the route to the 75% freight growth ambition by 2050.

Source	Tier	Date	Key claim used
Sensors (MDPI) , Bris-Peñalver , Verdecia-Peña and Alonso	Tier 2	Jan 2026	Surveying machine-learning methods applied to track geometry, vibration monitoring and imaging inspection, the authors find AI predictive maintenance for rail is held back by fragmented datasets and inconsistent measurement standards, poor cross-network generalisation, lack of interpretability in a safety-critical regulatory context, and disconnection between predictive models and real-time asset management platforms.
Railway Gazette International	Tier 3	Sep 2026	Alstom has won a £1bn contract, £800m rolling stock and £200m maintenance over eight years, for 29 five-car Adessia Stream battery-electric multiple units for TransPennine Express, with charging infrastructure at Hull, Scarborough and Saltburn, fleet delivery from 2032 and entry into service in winter 2034.
Économie Matin (<i>in French</i>)	Tier 3	Sep 2026	Alstom has signed contracts worth 1.2 billion euros with TransPennine Express for 29 battery-electric trainsets to be built at the historic Derby Litchurch Lane works, entering service from 2032 on routes in the north of England, using the Adessia Stream platform already deployed in Germany and France.
Rail Engineer	Tier 3	Apr 2026	ECDP has moved the Northern City Line to full ETCS with lineside signals removed and upgraded Welwyn to Hitchin, with Class 717 and 700 units fitted, Class 800 tested and Class 387 fitment underway, but freight locomotive fitment, above all the Class 66 fleet, is the major obstacle, with work on other freight fleets paused and DfT funding extending only through Railway Configuration State 3.

Source	Tier	Date	Key claim used
International Railway Journal	Tier 3	Apr 2026	GBRX, the innovation arm led by Toufic Machnouk, identifies three barriers to technology adoption on Britain's railway: commercial fragmentation, contractual and regulatory barriers designed into the Network Code, safety validation regime and IPR rules, and insufficient in-house expertise in frontier technologies, and intends to accelerate AI, quantum positioning, computer vision and sensor deployment.
Alstom	Tier 4	Sep 2026	Alstom states it has signed contracts worth 1.2 billion euros with TransPennine Express for 29 five-car battery-electric multiple units, to be built at Derby Litchurch Lane Works with manufacturing starting in 2028 and delivery from 2032, supporting over 350 skilled jobs in Derby and around 5,500 roles across the UK supply chain.

Theme 3: Maritime, where Britain wrote the rulebook for a fleet it does not build

Source	Tier	Date	Key claim used
Department for Transport (GOV.UK)	Tier 1	Aug 2026	Respondents reported grid connection lead times of up to 15 years, with one port told connection would not occur until 2039, and shore power standing charges of 157,093 to 413,749 pounds a year, with transmission costs rising from 47,045 to 120,289 pounds annually between 1.8MW and 2MW; shore power itself is described as mature and proven.
Maritime and Coastguard Agency (GOV.UK)	Tier 1	Jul 2026	The IMO published the non-mandatory MASS Code on 1 July 2026, the first international regulatory framework for remotely operated and autonomous cargo ships, with the MCA having led the drafting at the IMO, and a two-year experience-building phase now feeding into a mandatory version scheduled for 2032.

Source	Tier	Date	Key claim used
Department for Transport / Maritime and Coastguard Agency (GOV.UK)	Tier 1	Jun 2026	The UK government reports that a UK delegation was instrumental in developing the IMO MASS Code, described as the first international regulatory framework for remotely operated and autonomous cargo ships covered under SOLAS Chapter I.
Klima- og miljødepartementet (Norwegian Ministry of Climate and Environment) (in Norwegian)	Tier 1	May 2026	Norway will require every company operating on the Norwegian continental shelf to cut the greenhouse gas intensity of the vessels it uses in oil and gas activity from 2029, phasing in from a 10% reduction in 2029 to 2031 to 40% by 2038 to 2040.
Maritime and Coastguard Agency (GOV.UK)	Tier 1	Feb 2026	Workboat Code Edition 3 consolidates the UK's fragmented small commercial vessel codes into a single publication that includes what the MCA calls the world's first maritime safety legislation for remotely operated unmanned vessels and for workboats using alternative fuels, with the first cohort required to transition by 13 December 2026.
Results in Engineering (Elsevier), Gil-Pereira, Borges Lopes and Teixeira	Tier 2	Jun 2026	Benchmarking seven European port and logistics digital twin initiatives finds convergence on decision-centric twins with predictive and prescriptive analytics and on modular service-oriented architectures, but persistent gaps in documented physical-to-digital synchronisation, model performance reporting, latency detail, and in data availability where private stakeholders decline to share.
Havenbedrijf Rotterdam (in Dutch)	Tier 2	Dec 2025	Rotterdam's 2050 port vision, prepared with the municipality and national government, describes a port running on green electricity, hydrogen and other largely non-fossil energy carriers, with cargo, transport and infrastructure digitally connected and a deliberate choice to move hydrogen and hydrogen carriers by pipeline wherever possible.

Source	Tier	Date	Key claim used
SAFETY4SEA, reporting DNV Alternative Fuels Insight	Tier 3	Apr 2026	DNV recorded 45 alternative-fuelled vessel orders in Q1 2026, a fall of roughly 40% from 73 in Q1 2025, while 60 alternative-fuelled vessels were delivered in the quarter, 27 LNG-fuelled and 17 methanol-fuelled, with ammonia and hydrogen newbuild ordering remaining marginal.
Hill Dickinson LLP	Tier 3	Oct 2025	The MEPC extraordinary session in October 2025 voted 57 to 49 with 21 abstentions to adjourn adoption of the IMO Net-Zero Framework for twelve months, with the US, Saudi Arabia and Russia opposing and the EU states and the UK supporting adoption; the framework is now unlikely to enter into force before 2028.

Theme 4: Aviation, where the fuel pathway became law before the fuel

Source	Tier	Date	Key claim used
UK Civil Aviation Authority	Tier 1	Jul 2026	The concept of operations sets a phased roadmap for integrating beyond-visual-line-of-sight unmanned aircraft into UK airspace across three operational environments, with an objective to enable routine integrated BVLOS operations by 2027, positioning uncrewed traffic management as a strategic mitigation under the UK SORA framework, structured as current segregated operations, then 2027 to 2028 multi-operator with reduced segregation, then full integration from 2028.
Department for Transport (GOV.UK)	Tier 1	Jun 2026	DfT is launching a 219 million pound low carbon fuels fund in summer 2026, with 93 million available immediately to develop and scale SAF production, on top of 198 million already committed through the Advanced Fuels Fund since 2022; the SAF Mandate starts at 2% in 2025, rising to 10% by 2030 and 22% by 2040.

Source	Tier	Date	Key claim used
Department for Transport (GOV.UK)	Tier 1	Jun 2026	DfT reopened the evidence base on SAF supply, restating that the mandate's power-to-liquid obligation comes into force in 2028, that HEFA-derived SAF falls as a proportion of eligible supply from 2027 to 2040, and that government remains supportive of diversifying SAF technologies.
The National Archives / UK Parliament (legislation.gov.uk)	Tier 1	Mar 2026	The Act empowers the Secretary of State to direct a designated counterparty to offer revenue certainty contracts to UK sustainable aviation fuel producers, under which the counterparty pays the producer when the strike price exceeds the market reference price and the producer pays back when the reference price is higher; the direction-making power expires ten years after Royal Assent unless extended.
UK Civil Aviation Authority	Tier 1	Sep 2025	The CAA's eVTOL Delivery Model sets out the regulatory pathway covering design certification, airworthiness standards and pilot licensing, committing the regulator to put in place the frameworks and operational systems that allow initial commercial passenger eVTOL flights in the UK by the end of 2028, with a phased roadmap spanning aerodromes, air traffic control, airspace, security, flight training and manufacturing oversight.
UK Civil Aviation Authority	Tier 1	Evergreen	UKADS is a single guiding mind to coordinate and sponsor future airspace changes, replacing a model in which multiple organisations sponsored individual changes; it is delivered by NATS En Route plc, funded by a new Airspace Design Charge on airspace users, initially focused on London, with the CAA having modified the NERL licence on 25 March 2026 with effect from 6 May 2026.

Source	Tier	Date	Key claim used
Sustainable Aviation (UK industry coalition)	Tier 2	Jul 2026	The central scenario has SAF reaching 65% of total aviation fuel use by 2050 contributing roughly 31% of total decarbonisation, against a UK mandate requiring 22% by 2040; current-generation aircraft deliver around 10% of decarbonisation by 2050 and next-generation aircraft offering 25% efficiency improvements a further 10%, while entry into service for hydrogen narrow-body aircraft has been pushed later than the coalition assumed in 2023.
Deutsches Zentrum für Luft- und Raumfahrt (DLR) (in German)	Tier 2	Mar 2026	DLR reports testing prototype hydrogen pump and thermal management hardware at its newly built Future Propulsion Test facility, working on the distribution and conditioning systems needed to carry liquid hydrogen, which only liquefies below minus 253 degrees Celsius, from aircraft tank to engine combustion chamber.
Aviation International News / FutureFlight	Tier 3	Apr 2026	Vertical Aerospace's VX4 prototype completed a full two-way piloted transition flight, moving under pilot control between vertical thrustborne and horizontal wingborne flight and back, at the company's Kemble facility in Gloucestershire, making the UK developer the second in the world to accomplish the milestone.
Runway Girl Network	Tier 3	Feb 2026	ZeroAvia has pushed full ZA600 certification back by roughly two years and now targets certification of the fuel cell system in 2027, having previously aimed for initial powertrain certification by end-2025; funding constraints forced the company to halve its workforce.

Theme 5: The assurance and data spine beneath all four modes

Source	Tier	Date	Key claim used
Department for Transport (GOV.UK)	Tier 1	Jul 2026	DfT's stated research priorities include questions on emerging transport modes, on autonomous and remotely operated transport technologies, and on where current and future workforce pressures are likely to emerge, alongside a statement that better use of data will push forward research in all identified themes.
EASA (European Union Aviation Safety Agency)	Tier 1	Jul 2026	EASA's safety information bulletin, aimed at competent authorities, air traffic management and navigation service providers, air operators and manufacturers, highlights the growing severity and sophistication of GNSS jamming and spoofing incidents.
Department for Transport (GOV.UK)	Tier 1	Jun 2026	The consultation sets the statutory safety standard that authorised self-driving vehicles must achieve a level of safety equivalent to or higher than that of careful and competent human drivers, and states that the Secretary of State must have particular regard to the Statement of Safety Principles when deciding at authorisation whether a vehicle can travel safely and autonomously.
ENISA (European Union Agency for Cybersecurity)	Tier 1	Jun 2026	ENISA's biennial pan-European cyber crisis exercise for 2026 used a scenario in which critical maritime and railway infrastructures across Europe were simultaneously targeted in a coordinated cyberattack, and ran with over 100 leading cybersecurity experts and over 5,000 participants from national agencies and the public and private sectors.

Source	Tier	Date	Key claim used
House of Commons Library	Tier 1	May 2026	The briefing records that organisations in specified critical sectors including transport have statutory cyber security responsibilities under the Network and Information Systems Regulations 2018, with each sector overseen by a competent authority responsible for guidance, monitoring and enforcement, and sets out why the Bill widens those duties and regulator powers.
House of Commons Transport Committee	Tier 1	May 2026	The report and government response record that only 7% of total science graduates entered the manufacturing workforce in academic year 2021/22 and that apprenticeship starts for people under 25 have declined by 40% over the last decade across all sectors, with government pointing to Technical Excellence Colleges and apprenticeship reform as the response.
Department for Transport	Tier 1	Apr 2026	DfT commits to creating a Transport Data Marketplace as a single port of call for transport data, to promoting APIs as the expected method of sharing transport data, and to developing a Transport Data Ontology as a structured way of defining and linking the many parts of the transport system.
Department for Transport	Tier 1	Apr 2026	The strategy commits that by 2030 government will make available the essential tools for major city regions to roll out integrated contactless ticketing locally in a way that suits their needs, alongside 616 million pounds for walking, wheeling and cycling infrastructure and a commitment to open up operator data to the private sector.

Source	Tier	Date	Key claim used
Inside GNSS	Tier 3	Nov 2025	The UK government announced a 155 million pound positioning, navigation and timing package split across four initiatives, reported as roughly 71 million for an eLoran programme, 68 million for a National Timing Centre, 13 million for GNSS interference monitoring and 3 million for space-based time transfer research, led through the National PNT Office within DSIT.

Claim-fidelity self-disclosure

This briefing rests on 49 sources, every one of which was opened and read rather than taken from a search result. 44 are in English, 2 in German and one each in French, Norwegian and Dutch. Ten spans are quoted word for word; the rest are summarised. Where we summarise, we have tried not to make a source say more than it says. This box records the places where that discipline was hardest.

A claim that rests on an absence

The aviation theme's sharpest finding is that no government source publishes how many UK sustainable aviation fuel plants are under construction. An absence is weaker evidence than a presence: we can show that the June 2026 funding announcement names two projects and gives no total ([DfT](#), Jun 2026), but we cannot prove no such count exists anywhere. The claim should be read as what it is, that the number is not published where a reader would look for it, and the theme's counter-argument states the case for regarding this as premature rather than as failure.

Two readings carried from European sources

The cross-modal theme uses the European cyber exercise ([ENISA](#), Jun 2026) and the European aviation regulator's satellite navigation bulletin ([EASA](#), Jul 2026) as read-across rather than as UK regulatory evidence. Neither body regulates the UK. The cyber scenario quoted is a simulated exercise, not a record of an actual attack, and is described that way wherever it appears. We use both for the shape of the risk and not for any UK-specific obligation.

The weakest-evidenced part of the road theme

No current UK-specific source on vehicle-to-everything communication could be verified inside the recency window at any tier. The connected half of connected and automated vehicles is therefore carried by a single peer-reviewed paper ([Case Studies on Transport Policy](#), Feb 2026), which addresses it only lightly. Readers should treat the connectivity element of the road theme as less evidenced than the autonomy and battery elements beside it.

Sources dropped rather than stretched

Four sources were gathered, read and then excluded. A 2023 vendor release on the first ammonia-powered container ship and a 2024 report on green shipping corridors both sat outside the recency window and are not foundational legislation or long-run statistics, so neither could honestly be carried as a structural anchor; the ammonia point is instead made from Q1 2026 orderbook data ([DNV's Alternative Fuels Insight, reported by SAFETY4SEA](#), Apr 2026) and the green-corridor claim is not made at all. A July 2025 law-firm note stating that the five-plant SAF target would be missed was dropped on the same test, which is part of why the plant-count finding above rests on an absence. An international briefing on the autonomous ships code was dropped because the page carries no publication date, and the dated UK government announcements are used instead ([the Maritime and Coastguard Agency](#), Jul 2026) ([DfT](#), Jun 2026).

Three quotes the machine could not check

Every quoted span in this briefing was fetched against its live page by the provenance gate. Three could not be, and all three are the non-English ones, which is not a coincidence: the gate's HTTP client is the weakest link on European government sites. The German ministry page returns its content in a form the gate cannot parse, the French trade page fails the TLS handshake outright, and the Norwegian ministry returns 403 to the gate's client. Each of the three was therefore verified by hand against the rendered page on 17 September 2026, and the sentence on either side of each quote is recorded in the source file so a later reader can check the verification rather than take it on trust. The quotes themselves were not altered. This is the one place in this cycle where the reader is relying on an analyst rather than on a gate, and it is worth knowing which three ([the German federal transport ministry](#), Jul 2026) ([Economie Matin](#), Sep 2026) ([the Norwegian Ministry of Climate and Environment](#), May 2026).

The recency window is twelve months, not six

The house default is a soft six-month window. This cycle declares twelve, and the difference is material enough to state. Twelve of the 49 sources sit between six and twelve months old, and they carry load: the gigafactory capacity figures ([the UK Gigafactory Commission](#), Jan 2026), the regulator's eVTOL delivery model ([CAA CAP 3169](#), Sep 2025), the workboat code ([the Maritime and Coastguard Agency](#), Feb 2026), the vote that adjourned the global shipping fuel regime ([Hill Dickinson](#), Oct 2025), the hydrogen certification slippage ([Runway Girl Network](#), Feb 2026), the

positioning resilience package ([Inside GNSS](#), Nov 2025) and the peer-reviewed assessment of UK assurance capability ([Case Studies on Transport Policy](#), Feb 2026). On a brief whose horizon is 2040, a six-month window would have excluded the most structural evidence in the base while admitting whatever happened to be published in the last quarter. We widened the window rather than reclassify news items as structural anchors to get them past the gate, which is the other way this problem is usually solved and is not honest. The audit for this cycle must therefore be run with a 365-day window; at the house default it reports twelve failures, and those twelve are this paragraph.

A rule that foundational legislation cannot satisfy

Two Acts are carried as structural anchors ([the Automated Vehicles Act 2024](#), May 2024) ([the Sustainable Aviation Fuel Act 2026](#), Mar 2026). The sourcing guide says a structural anchor should not be more than twelve months old, which primary legislation almost never is, and the audit duly warns that the Automated Vehicles Act is 850 days old. We have not suppressed the warning. It is a conflict in the rule rather than a defect in the source, and it is recorded here so that the next cycle inherits the question rather than the workaround.

Dates recorded to the month

Six documents print only a month and year on their title page. In each case the first of that month is recorded and the absence of a day is noted in the source record, rather than a day being inferred from a web address or a file path. One regulator page carries no date at all and is recorded as undated ([the CAA](#), undated).

One figure the previous cycle asserted and this one withdraws

The May 2026 cycle tracked a decision expected at a 2026 spending review. There was no 2026 spending review. The continuity ledger grades that reading as overturned rather than dropping it, because a milestone that cannot be tested as written is a finding about the previous analysis and not an administrative detail.

This Decision Intelligence briefing is the quarterly synthesis of a Shaping Tomorrow intelligence cycle on the **UK Department for Transport**. It sits on top of weekly Signal Scans that identify emerging developments and weak signals, and monthly Change Trackers that assess what is materially shifting in momentum. Cycles may also close with Action Triggers, an end-of-cycle synthesis that reads across all four formats and surfaces patterns no single deliverable carries on its own. Continuous intelligence on this topic is available as a Single-Topic or Multi-Topic programme.

Discuss a Tailored Programme

Prepared by Shaping Tomorrow: 17 September 2026